Document title	Fix consumer lag on the orders queue
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Revision	01
Effective date	2024-11-02
Owner	Orders team
Services	orders-worker, order-queue
Alerts	OrderQueueConsumerLagHigh
Supersedes	None

Standard operating procedure

Fix consumer lag on the orders queue

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Appendix A. Revision history

1. Purpose

Clear a growing backlog on the orders topic so orders are processed again.

2. Scope

Applies to orders-worker in the shop-prod cluster.

3. When to use

Use this procedure when the OrderQueueConsumerLagHigh alert fires, or when orders stay in pending for more than a few minutes.

Alerts: OrderQueueConsumerLagHigh

4. Definitions and abbreviations

- Consumer lag: The number of messages on the orders topic that the orders-worker group has not processed yet.

5. Roles and responsibilities

Role	Responsibility
On-call engineer	Runs the procedure.

6. Related documents

None.

7. Prerequisites

- kubectl access to the shop namespace.

8. Procedure overview

Check the lag, restart the workers, then check the lag again.

9. Procedure

Run the steps in order. Do not skip or reorder steps. Stop and escalate if a result does not match the expected result.

Step	Action	Command	Expected result	Responsibility	Automation
1	Check the consumer lag.	`kafka-consumer-groups --bootstrap-server order-queue:9092 --describe --group orders-worker`	Lag above 10000 on one or more partitions.	On-call engineer	check:queue_lag
2	Restart the workers.	`kubectl -n shop rollout restart deploy/orders-worker`	All worker pods restart and rejoin the consumer group.	On-call engineer	action:restart_worker
3	Check the lag again after five minutes.	`kafka-consumer-groups --bootstrap-server order-queue:9092 --describe --group orders-worker`	The lag is going down.	On-call engineer	check:queue_lag

10. Verification

- Lag is under 1000 on every partition.

11. Rollback

None.

12. Troubleshooting

None.

13. Escalation

- Page the orders team if the lag keeps growing after the restart.

14. Security considerations

None.

15. Compliance and standards

None.

16. Review and update

Reviewed every year.

17. Records

Note the restart in the incident ticket.

Appendix A. Revision history

Revision	Date	Author	Change
01	2024-11-02	Luis Romero	First version.